

The Making of Dominant Vehicle Currencies: Evidence from DeFi

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Abstract

We reconstruct 475 million swaps and observe routes' vehicle currencies. From 2024 H1 to 2026 H1, stablecoin share rises from 16.9% to 42.4%. In percentage pointpercentage points (pppp), net switching within continuing pairs is -0.1 pp, while pairs first observed after 2024 H1 contribute $+20.1$ pp of the $+25.5$ pp increase. Among pairs that trade again, entry vehicle identity predicts use through day 120. Holding pair, date, input, and pretrade state fixed, the stablecoin price-leadership coefficient is $+57.59$ pp. Among mature pairs, lagged full-range constant-product capital on the weaker legs predicts retention beyond price, while retained incumbents leave 3.3 bp of input-value-weighted output. Pair entry and activity reallocation drive aggregate rotation; prices and two-leg capital shape persistence.

1 Introduction

A vehicle currency can dominate trading even when no party wants to hold it. It is acquired only to be exchanged again because two liquid markets through the vehicle can outperform a thin direct market. The U.S. dollar appears on more than 85% of foreign-exchange transactions, far more often than the United States' share of cross-border economic activity would suggest (Somogyi, 2026). Vehicle use concentrates trading in dollar markets; deeper markets lower trading costs and reinforce the dollar's intermediary role (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). This feedback explains persistence after a vehicle becomes dominant. It says less about how a challenger gains share as new trading relationships and market technologies appear.

Aggregate dominance combines two margins. A continuing relationship can replace its vehicle, or activity can move toward relationships that already use another vehicle and toward new relationships formed around it. The same aggregate change can therefore arise with pervasive switching or with no net switching inside established relationships. Separating these cases requires transactions that attach the intermediary to the exchange between its source and destination.

Standard foreign-exchange records report turnover in each currency pair without linking the legs of the customer's exchange.¹ A yen-euro exchange routed through dollars appears as separate

¹For example, see the [BIS 2025 Triennial Survey reporting guidelines and templates](#).

yen-dollar and dollar-euro trades. Somogyi (2026) infers vehicle activity from currency-pair volume responses around non-overlapping holidays. In decentralised exchange, the pool calls inside one blockchain transaction preserve their execution order. We use those records to observe the connected route directly.

We call the ordered source and destination the *endpoint pair*, shortened hereafter to *pair*. Each directed pool execution is a *leg*; the ordered sequence of connected legs is a *route*; and an asset between the two endpoints is a vehicle. We reconstruct routes from 475 million swaps on 9 Ethereum automated market maker (AMM) deployments from November 2018–June 2026; Table 1 reports the resulting panel. The token sequences identify the pair, every intermediary, and whether the route spans exchanges. This transaction-level object reveals vehicle use without inferring a connection from turnover in separate markets.

The sample begins with an institutional benchmark for how a vehicle network forms. Uniswap v1 allowed only ETH-token pools, so each token-to-token trade had to cross ETH; we recover 217,003 such routes. Uniswap v2 removed that rule and allowed any two ERC-20 tokens to form a pool. Six years later, however, 95.5% of single-leg v2 trades still use a WETH pool and 97.9% of token combinations first traded in 2026 include WETH. Appendix Table 8, panels A and B, reports these institutional facts. Removing the mandate did not unwind the pool network formed around the native asset. We read this chronology descriptively because venue migration and available token combinations changed with the rule.

Stablecoins, including USDC, USDT, DAI, and other pegged tokens, become substantially more prominent vehicles later in the sample. Giving each matched January–June day equal weight, their share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value between 2024 and 2026 (Appendix Table 18). The half-year series in Figure 1 reverses direction: native assets regain share in 2023 and 2024 before stablecoins recover the routed-value lead in 2025. A smooth technology trend is therefore an incomplete explanation.

We decompose the pooled route activity into those margins. The route-count share rises from 16.9% to 42.4%, or +25.5 pp. Moves toward and away from stablecoins within continuing pairs net to −0.1 pp, and activity shifting across continuing pairs contributes +8.4 pp. The +17.8 pp contribution from period-specific vehicle activity is a net: pairs first observed after 2024 H1 contribute +20.1 pp, while reactivation, vehicle-role turnover, and pair exit together contribute −2.3 pp. First-observed entry alone supplies 78.9% of the route-count rotation. Panel B of Table 2 gives the dollar-weighted counterpart: first-observed entry contributes +21.9 pp, or 51.2% of the total change, before the other lifecycle margins reduce the period-specific net to +19.2 pp. We therefore locate the aggregate rotation in pair entry and the distribution of trading across continuing pairs; vehicle changes inside continuing pairs largely offset in aggregate.

The vehicle used at pair entry predicts later allocation. Among pairs that trade again, a 10 pp higher entry-day stablecoin share predicts +8.92 pp higher stablecoin share during days 1–30 and +8.40 pp during days 31–120 (Table 3, panel A, columns 2 and 5). The entry share also predicts a smaller increase in the probability that the pair trades again (panel B). The disjoint windows

exclude the entry day and separate persistence in vehicle identity from pair survival.

Depth on both route legs provides the corresponding market structure. A stablecoin bridge often becomes technically available while remaining shallow. First-month adoption is 42.6% when its weaker leg has less than one-tenth of WETH depth and 84.1% once it reaches that threshold (Table 4, panel C). Among eventual adopters, deposited capital on the weaker leg rises before the first observed stablecoin route (Figure 2). In the exact stablecoin-versus-WETH comparison, the coefficient on stablecoin price leadership is +57.59 pp and the incumbent-retention coefficient on challenger price leadership is −58.08 pp (Table 5, columns 1 and 2). In the joint model, the coefficients on a 100-basis-point incumbent output advantage and a 10 pp incumbent lagged full-range constant-product capital advantage are +10.13 pp and +2.77 pp, respectively (column 3). Their interaction is −0.44 pp (column 4), consistent with lagged capital carrying a stronger retention association when the incumbent quote is weaker.

We contribute an extensive-margin account of currency dominance and evidence on the conditions that sustain an incumbent vehicle. The route panel complements the holiday-based inference of Somogyi (2026) and the historical network evidence of Flandreau and Jobst (2009). The pair decomposition shows how aggregate leadership can rotate while vehicle changes inside continuing pairs offset. The post-entry, bridge-depth, and exact-price estimates then connect formation to classical volume–cost feedback, constrained currency-market liquidity supply, and decentralised-exchange liquidity provision (Krugman, 1980; Huang et al., 2025; Lehar and Parlour, 2025). The price comparison extends optimal-routing work from parallel pools to sequential routes through alternative vehicles (Xi and Moallemi, 2026). Together, the results show where a vehicle network expands, when an incumbent survives a cheaper challenger, and how current relative prices and lagged two-leg capital divide that contest.

The extensive margin is also relevant for payment-system design. Project Nexus assigns currency conversion to a foreign-exchange provider, while Project Rialto uses a third currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2024, 2025). Access policy can therefore affect the price inside an existing relationship and which two-sided markets become the bridge inherited by new ones.

2 Setting and data

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine atomically. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Because the pool calls share a transaction, the intermediary sequence is observable.² A route through an intermediary is unavoidable when no direct pool connects the assets; when direct and indirect pools coexist, the router compares their fees, price impact, gas,

²Ethereum records a transaction’s ordered logs in its receipt; see the official [Ethereum JSON-RPC documentation](#).

and venue access. [Lehar and Parlour \(2025\)](#) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction. For a reconstructed sequence $A \rightarrow B \rightarrow C$, (A, C) is the pair, $A \rightarrow B$ and $B \rightarrow C$ are its legs, and the full sequence is the route. All three objects are directed by token flow. Each leg executes in a liquidity pool; a pair need not have a direct pool. The observed route endpoints belong to the connected pool component and need not coincide with the endpoints of a broader user instruction.

A transaction on January 10, 2026 illustrates both the information in the logs and its limit. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.³ The user’s instruction begins with PYUSD, but the connected pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The observed pair is therefore USDC $\rightarrow$ USDe; its legs are USDC $\rightarrow$ USDT and USDT $\rightarrow$ USDe, and USDT is the vehicle. The earlier PYUSD $\rightarrow$ USDC conversion occurs outside the connected pool sequence. The data do not show whether USDC was another vehicle or inventory supplied by the executor. We apply the connected-component rule to every transaction and do not infer an unobserved leg from nearby transfers. This choice may understate the intermediary role of assets such as USDC when a broader instruction is funded outside the observed pool sequence.

The sequence directly identifies intermediary use. Conditional on reconstructed pool states immediately before execution, it also defines a same-size cost comparison between the observed route and feasible alternative paths. We introduce notation for the route here and develop the cost comparison in Section 5.

For a route r , let $e(r)$ record its block, transaction position, and first pool-event position; $e(r)^-$ is the state immediately before that transaction begins. A route carries an ordered pool sequence $(p_1, \dots, p_{m(r)})$ and token sequence $(k_0, \dots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p ’s pretrade state and let $Q_p(i \rightarrow j, \Delta_i; \chi_{p,e^-})$ denote its exact-input output in token j . Composing the pool quotes gives the route’s token output and dollar value,

$$\begin{aligned} \Delta_{k_h} &= Q_{p_h}(k_{h-1} \rightarrow k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}), & h &= 1, \dots, m(r), \\ Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o, e(r)} Q_r(q, e(r)). \end{aligned} \tag{1}$$

The venue-specific forms of Q_p appear in Appendix B. Pool invariants take token quantities as arguments, which is why the dollar notional q and the token input Δ_i remain distinct. Equation (1) describes an executed route and, once market state has been reconstructed, prices a counterfactual path.

Intermediary use is one of several currency roles. Invoicing concerns the currency named in

³Transaction [0xbda1d07f...9bf67ad9](#).

trade contracts (Gopinath et al., 2020; Amiti et al., 2022; Mukhin, 2022); funding concerns the denomination of liabilities (Gopinath and Stein, 2021; Eren and Malamud, 2022); and dealer turnover records trading activity (Somogyi, 2026). These roles can reinforce one another. The route in Equation (1) identifies the asset used between the currency sold and the currency bought.

This route setting makes the classical volume–cost mechanism observable at transaction level. Greater use of a vehicle expands trading in the pools supporting its two legs; deeper pools can then lower the cost of using the same vehicle again (Krugman, 1980). In Kiyotaki and Wright (1989), traders accept an asset for indirect exchange when its marketability compensates for its storage cost. Dowd and Greenaway (1993) emphasise the fixed costs of learning, accounting, and quoting in a new currency. Liquidity placement, venue access, and the routes available to software provide corresponding coordination forces on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route consists of one pool trade on either side of a unique intermediary and identifies one mutually exclusive vehicle choice for each endpoint exchange. A longer route may contain both a native asset and a stablecoin. We retain those routes in two complementary measures: each intermediary position contributes to the full-route share, and route participation records whether a currency appears anywhere inside a route. Section 4.1 shows that stablecoins gain in both the two-leg choice sample and routes with more than two legs. For route r , let $\mathcal{M}(r)$ denote the set of assets between its source and destination. We record the intermediary status of asset k as

$$Z_{r,k} = \mathbf{1}_{\{k \in \mathcal{M}(r)\}}.$$

Let $a(k)$ assign each token to an economic asset type. The native group consolidates ETH and its one-for-one wrapped representation, WETH, after route reconstruction. The stable group contains fiat-reserve, on-chain-collateralised, synthetic, and non-dollar stable currencies. For $g \in \{\text{native, stable}\}$, define $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$; on an exact two-leg route, this sum is an indicator.

This grouping preserves a token-level estimand. WETH is the one-for-one token representation that allows ETH to interact with ERC-20 pool contracts; wrapping changes the technical form, not the underlying settlement exposure. We therefore reconstruct routes at the token level and consolidate ETH and WETH only when assigning the economic asset used as a vehicle. Liquid-staking tokens remain separate because they also convey staking returns. Dollar-pegged tokens remain distinct instruments because their issuers, backing, redemption arrangements, risks, and leg-level liquidity differ (Gorton and Zhang, 2023; Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). USDT in the USDC–USDT–USDe example is therefore an intermediary token even though all three instruments target the same unit of account. The aggregate stable group measures dominance by a family of stable tokens, while the token-level results retain issuer-specific competition.

Native assets and stablecoins define the main comparison because they are the two persistent broad vehicle families in Figure 1. Staked-native, imported, and other intermediaries remain in the wider time series and supplementary results; they do not enter the native-versus-stable denominator. For routes with more than one intermediary, realised intermediary counts retain each asset used along the route. Appendix Table 22 also compares realised use with graph betweenness. Betweenness records how often an asset could lie on paths in the observed trading graph; route shares record the intermediary currencies traders actually use.

Let $\mathcal{R}_t^{(2),NS}$ denote exact two-leg routes on day t whose intermediary belongs to either the native or stable group, and let $\mathcal{R}_t^{(2),NS,V}$ retain routes for which the source, intermediary, and destination values agree within 20%. The equally weighted and dollar-weighted group shares are

$$S_{g,t}^N = \frac{\sum_{r \in \mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \quad S_{g,t}^V = \frac{\sum_{r \in \mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r \in \mathcal{R}_t^{(2),NS,V}} \nu_r}, \quad (2)$$

where ν_r is the arithmetic mean of the route’s total dollar value at its observed source and destination. The 20% rule also requires every intermediary’s incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. Thus $S_{stable,t}^N$ is stablecoins’ aggregate share among native and stable intermediaries, and $S_{stable,t}^V$ is the corresponding dollar-weighted share. The two groups exhaust this denominator. Share levels are reported in percent; the unit for absolute share changes is one pp.

Appendix G reports two complementary comparisons: all-route participation and intermediary use relative to endpoint demand. It also examines the two largest reserve-backed stablecoins separately. These measures retain longer routes and the broader set of native, staked-native, stable, and imported currencies; Equation (2) remains the measure used for the aggregate native-versus-stable rotation.

2.3 The route panel

The principal route panel covers Curve, Uniswap v1 through v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. We selected these exchange families for their historical Ethereum volume and retained the deployments whose transaction-level pool events and token identities can be reconstructed consistently. The panel spans 2,798 dates from November 2018–June 2026 and covers major constant-product, concentrated-liquidity, stable-swap, weighted-pool, singleton, and integrated-liquidity designs. Other Ethereum exchanges remain outside it. Appendix Table 15 reports both volume shares within the retained source series and the selected families’ share of total Ethereum DEX volume in DeFiLlama.

Indexed pool records or factory registries usually link a pool to its token contracts. Uniswap v1 records the exchange contract in each transaction, and its immutable exchange registry maps every observed contract to a token address and symbol. Shared transaction hashes and matched ETH amounts then identify the ordered legs of v1 token-to-token routes. Section 4.2 uses these records

to study the protocol rule that made ETH intermediation compulsory.

The source data contain 475,071,108 pool-level swap records. Indexed blockchain data supply events for eight exchange series, Dune’s `dex.trades` table supplies Fluid records, and direct Ethereum queries supply block headers, ordered logs and receipts, token precision, factory registries, and state checks.⁴ After identifying trade direction and pool events and standardising token units, 474,388,425 directed pool-swap legs remain. We order the legs within each transaction and apply the flow rule above to distinguish single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction algorithm and reports the historical token units that we cannot verify.

Transactions with more than one disconnected pool-event group pose a separate measurement limit: the observed calls do not reveal whether the groups belong to one broader instruction (Xi and Moallemi, 2026). The principal sample excludes every component in those transactions. Appendix Table 7 treats each internally valid component as a separate route and finds a slightly larger stablecoin rotation.

Every non-self-returning token sequence retained by the connected-transaction rule enters the count measure. Dollar-weighted results require source, intermediary, and destination values to agree within 20%. This is an eligibility test for a coherent dollar notional, not a control for route characteristics. Keeping the count sample broader preserves observed intermediary choice when valuation is noisy; Appendix F reports the separate round-trip exclusion, and each value exhibit reports the stricter sample’s coverage.

Transactions and contract-state changes are publicly observable (Schär, 2021). Reconstructing a route still requires matching pool events, venues, and token units. This procedure identifies the executed route and its calling contract, while the executor address does not establish which frontend or software formed the quote (Xi and Moallemi, 2026). The results are therefore attributed to executed routes, not to a particular routing service.

Table 1 reports the resulting panel coverage and route frequencies. Appendix A gives the route-construction checks, while Appendix B validates the pretrade pricing models used for the counterfactual comparisons.

⁴See [The Graph’s indexing documentation](#), [Dune’s curated-data documentation](#), and the official [Ethereum node-interface documentation](#).

Table 1: The route panel

	Route panel
Dates in the full route panel	2,798
Missing source-days	0
Raw swap rows	475,071,108
Usable directed route legs	474,388,425
Routes per day, median across eligible days	153,586
Multi-leg routes per day, median across eligible days	30,064
Multi-leg share of routes, median across eligible days	19.4%
Round trips, median share of multi-leg routes by count	12.8%
Round trips, median share of multi-leg routes by value	21.0%

Note: The full 9-deployment panel contains UTC dates from November 2018–June 2026. Uniswap v1 enters these totals through the exact exchange-to-token registry; its protocol-architecture comparison is venue-specific. Daily route statistics and round-trip shares use all 2,449 eligible dates from 15 May 2019 through 30 June 2026. A multi-leg route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-leg routes.

3 How stablecoins gain the vehicle role

3.1 The aggregate rotation

Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. The comparison uses the same January–June dates in both years because the 2026 panel ends on June 30. Giving each matched day equal weight, stablecoins’ share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value. Appendix Table 18 reports the paired-day estimates and their sampling uncertainty. The decomposition below pools route activity so its components allocate the aggregate count or value exactly.

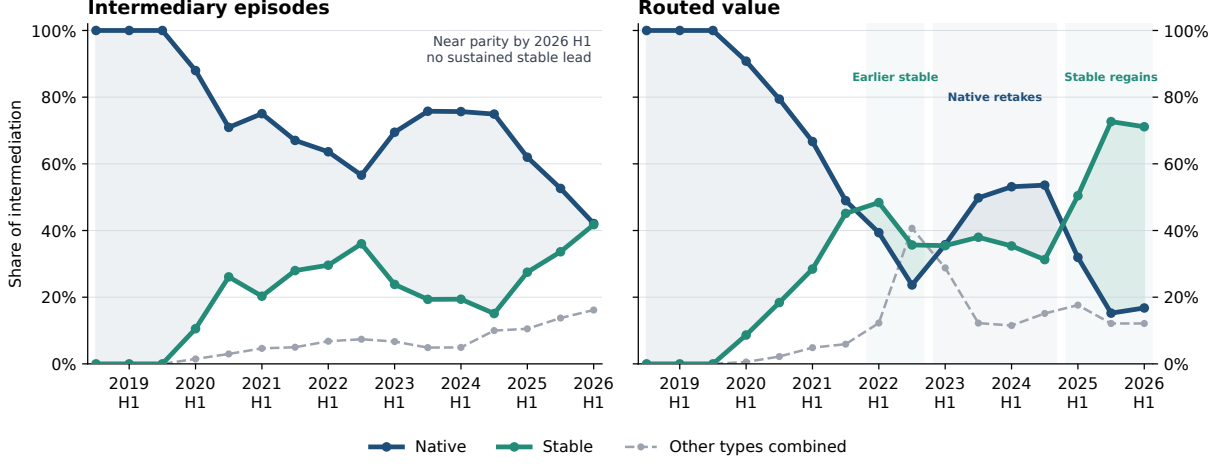
The half-year series in Figure 1 reverses direction. Native assets regain share in 2023 and 2024; stablecoins recover the routed-value lead in 2025. The issuer results below distinguish USDT and USDC within that family.

3.2 Pair formation and persistence

Aggregate dominance can change inside continuing pairs or through the distribution of activity across pairs. Let p denote a pair and $y \in \{2024, 2026\}$. A continuing pair carries native-or-stable routes in both matched periods; a year-specific pair carries them in only one. First split total activity between these two sets. Within continuing pairs, stablecoin share can change because a given pair changes its vehicle mix or because trading shifts toward pairs with a different vehicle mix. The remaining change comes from the aggregate weight on continuing pairs and the stablecoin share of pairs observed in only one period.

Formally, let W_y be the share of native-plus-stable activity in continuing pairs, $\omega_{p,y}$ pair p ’s weight within that activity, and $s_{p,y}$ its stablecoin share. Continuing pairs have stablecoin share $S_{C,y} = \sum_p \omega_{p,y} s_{p,y}$, while year-specific pairs have share $S_{E,y}$. Total stablecoin share is therefore

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete routes that do not return to their starting asset. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Appendix Table 18 uses the narrower native-or-stable denominator. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination dollar amounts to agree within 20%.

$S_y = W_y S_{C,y} + (1 - W_y) S_{E,y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$ gives the exact midpoint decomposition

$$\begin{aligned} \Delta S = & \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{activity shifting across pairs}} \\ & + \underbrace{(\bar{S}_C - \bar{S}_E) \Delta W}_{\text{continuing-pair weight}} + \underbrace{(1 - \bar{W}) \Delta S_E}_{\text{year-specific pairs}}. \end{aligned} \quad (3)$$

The midpoint averages hold the companion quantity fixed while each component changes; the four terms add exactly to the aggregate change. The first two terms split the change among continuing pairs into changes in each pair's stablecoin share and changes in pair weights. The last two terms separate the changing mass of continuing pairs from the stablecoin share among period-specific pairs. The two sets partition each period's activity; nothing is deducted from a continuing pair.

We also compare the same pair, date within the year, and realised single- or cross-exchange route class. For route-count or routed-value stablecoin share $S_{pds,y}^{(m)}$, the matched regression is

$$S_{pds,y}^{(m)} = \alpha_{pds}^{(m)} + \beta^{(m)} \mathbf{1}_{\{y=2026\}} + \varepsilon_{pds,y}^{(m)}, \quad m \in \{N, V\}. \quad (4)$$

Here d is the month-day and s the realised route class. Observations are weighted by native-plus-stable route count or supported routed value; standard errors cluster by pair and date. The comparison is descriptive because route class and trading activity can respond to the same forces

as vehicle choice.

Table 2 applies Equation (3) to route counts and supported value and reports Equation (4) on cells defined by pair, calendar date, and route class.

Pooling route counts gives an increase from 16.9% to 42.4%, or +25.5 pp, close to the equal-day change in Appendix Table 18. The within-pair component is -0.1 pp by route count and 0.0 pp by routed value. These net estimates combine +1.3 pp from pairs moving toward stablecoins and -1.4 pp from pairs moving toward the native asset. Activity shifting across continuing pairs contributes +8.4 pp by count and +26.2 pp by value. The changing aggregate weight on continuing pairs contributes -0.5 pp and -2.5 pp (Table 2, panels A and B).

The period-specific contribution is also a net. Pairs first observed after 2024 H1 contribute +20.1 pp by route count and +21.9 pp by routed value. Reactivation, vehicle-role turnover, and exit together contribute -2.3 pp and -2.8 pp, leaving the period-specific terms at +17.8 pp and +19.2 pp. The underlying rows add exactly before rounding. First-observed pair entry therefore supplies 78.9% of the total route-count rotation and 51.2% of the dollar-weighted rotation. Equation (4) reaches the same conclusion on the more restrictive matched sample: stablecoin share changes by +0.2 pp by count and -1.4 pp by value (Table 2, panel C). A large aggregate rotation therefore coexists with little net switching inside established cells defined by pair, calendar date, and route class.

The same decomposition covers every adjacent January–June comparison between 2019 and 2026. As shown in Appendix Table 20, panel A, the within-pair term ranges only from -0.4 pp to +1.0 pp, while aggregate changes range from -6.4 pp in 2022–2023 to +17.4 pp in 2025–2026. Pair composition therefore accounts for both stablecoin gains and the intervening reversals.

Endpoint demand clarifies the composition result. A pair with WETH at one endpoint cannot also use WETH as its intermediary, so part of the native-versus-stable allocation is mechanically fixed. Panel A of Appendix Table 21 splits the headline daily change by endpoint direction. Pairs with one native and one stable endpoint contribute +9.2 pp by route count and +10.1 pp by routed value; pairs with two stable endpoints contribute +1.0 pp and +13.2 pp, respectively. All other pairs remain the largest channel overall. Panel B shows that USDT supplies +13.7 pp of the two-stable-endpoint value increase, while USDC, DAI, and the remaining stablecoin intermediaries jointly contribute -0.5 pp. The result is present on all 181 common days, and trimming the largest decile of daily contributions leaves a +12.3 pp increase. Stablecoin dominance therefore spreads beyond pairs whose endpoint identity limits the native vehicle, while routing between stable currencies concentrates on one intermediary issuer.

Panel B of Appendix Table 20 removes every pair with WETH or a stablecoin at either endpoint. Stablecoin vehicle share still rises from 1.1% to 9.0% by count and from 0.9% to 39.1% by supported value. Pair entry and exit contribute +7.2 pp and +33.8 pp, respectively. The rotation therefore remains economically large among pairs for which both broad vehicle families are distinct from the assets being exchanged.

The first routes in a new pair predict its subsequent vehicle. For pair p , let E_p be the stablecoin

Table 2: Decomposition of the stablecoin rotation

Component or estimate	Estimate [pp]	Obs.
<i>Panel A. Route-count share: continuing and year-specific pairs</i>		
Net stablecoin-share change within continuing pairs	−0.1	
Pairs moving toward stablecoins (1,569)	+1.3	
Pairs moving toward native assets (1,487)	−1.4	
Vehicle activity shifting across continuing pairs	+8.4	
Weight of continuing versus year-specific pairs	−0.5	
Net contribution of period-specific vehicle activity	+17.79	
Pairs first observed after 2024 H1	+20.09	
Pairs reactivated after absence in 2024 H1	+0.20	
Vehicle-role turnover in continuing pairs	−0.77	
Pairs exiting before 2026 H1	−1.73	
Total route-count change	+25.5	
<i>Panel B. Dollar-weighted share: continuing and year-specific pairs</i>		
Net stablecoin-share change within continuing pairs	0.0	
Pairs moving toward stablecoins (1,505)	+2.3	
Pairs moving toward native assets (1,445)	−2.4	
Vehicle activity shifting across continuing pairs	+26.2	
Weight of continuing versus year-specific pairs	−2.5	
Net contribution of period-specific vehicle activity	+19.16	
Pairs first observed after 2024 H1	+21.94	
Pairs reactivated after absence in 2024 H1	+0.04	
Vehicle-role turnover in continuing pairs	−1.72	
Pairs exiting before 2026 H1	−1.10	
Total change in dollar-weighted share	+42.8	
<i>Panel C. Matched pair estimates</i>		
All two-leg routes, count share	+0.23 (0.77)	188,344
20% agreement sample, count share	+0.32 (0.75)	182,734
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,734

Note: Panels A and B compare pooled activity over matched January–June dates in 2024 and 2026 and report Equation (3). Their first indented rows split the within-pair component into positive and negative contributions. Their lifecycle rows split the net contribution of period-specific vehicle activity using each pair’s first and last observation in the full panel: first-observed entry, reactivation after absence in 2024 H1, vehicle-role turnover among pairs active in both periods, and exit before 2026 H1. These lifecycle rows add to the net term using unrounded values. Panel B weights routes by dollar value and applies the 20% agreement rule. Panel C estimates Equation (4); standard errors use two-way clustering by pair and date. Appendix Table 19 reports an alternative factorisation of the route-count change.

share on its entry day. For each of the disjoint windows $h \in \{1-30, 31-120\}$, let $R_{p,h}$ indicate that the pair carries another native- or stablecoin-intermediated route, and let $S_{p,h}$ be the stablecoin share when $R_{p,h} = 1$. We estimate

$$\begin{aligned} S_{p,h} &= \alpha_h + \beta_h E_p + X_p' \delta_h + \varepsilon_{p,h}, & R_{p,h} &= 1, \\ R_{p,h} &= a_h + b_h E_p + X_p' c_h + u_{p,h}, & \text{all entrant pairs.} \end{aligned} \tag{5}$$

The second line is a linear probability model. The controls X_p contain the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. The sample uses pairs entering by March 2 in 2024 or 2026, giving every pair 120 complete calendar days of follow-up. Pairs with a native asset at either endpoint are excluded.

We separate later vehicle use from pair survival. Of the 157,262 entrants, 19.4% trade again during days 1–30 and 12.3% do so during days 31–120 (Table 3, panel B). Among pairs that trade again, a 10 pp higher entry-day stablecoin share predicts +8.92 pp higher stablecoin share during days 1–30 and +8.40 pp during days 31–120 (panel A, columns 2 and 5). The route-activity-weighted estimates are +8.55 pp and +9.00 pp, respectively (columns 3 and 6). Every follow-up outcome begins after the entry day, so these estimates capture continued vehicle use across subsequent routes.

The share estimates describe pairs that remain active in each window. Entry-day stablecoin share also predicts a modestly higher probability of subsequent trading: a 10 pp increase corresponds to +0.30 pp during days 1–30 and +0.98 pp during days 31–120 (Table 3, panel B). The evidence therefore combines strong persistence in vehicle identity among surviving pairs with a smaller survival margin. We interpret both as descriptive equilibrium relationships: relative prices, depth, and venue access can shape the vehicle at entry and remain relevant for later trading.

Appendix G relates the formation result to network reach, issuer-level specialisation, within-day endpoint demand, and exchange span. Stablecoin arrival is more common in older and more active pairs, intermediary use moves closely with a currency’s endpoint demand within a day, and the stablecoin rotation appears in both single- and cross-exchange routes.

Table 3: Vehicle persistence after pair entry

<i>Panel A. Stablecoin route share among pairs that trade again</i>						
	Days 1–30			Days 31–120		
	(1)	(2)	(3)	(4)	(5)	(6)
Entry stablecoin share [effect pp per 10 pp]	+9.00*** (0.14)	+8.92*** (0.15)	+8.55*** (0.72)	+8.59*** (0.18)	+8.40*** (0.18)	+9.00*** (0.50)
Controls	No	Yes	Yes	No	Yes	Yes
Route-activity weights	No	No	Yes	No	No	Yes
Observations (pairs)	30,547	30,547	30,547	19,405	19,405	19,405
Entry-date clusters	123	123	123	123	123	123
Mean stablecoin share [pp]	2.70	2.70	4.78	3.83	3.83	7.18
R^2	0.842	0.843	0.800	0.787	0.791	0.830

<i>Panel B. Subsequent-trading incidence among all entrants</i>		
	Days 1–30	Days 31–120
Entry stablecoin share [effect pp per 10 pp]	+0.30*** (0.08)	+0.98*** (0.08)
Controls	Yes	Yes
Observations (pairs)	157,262	157,262
Entry-date clusters	123	123
Mean retrading rate [pp]	19.42	12.34
R^2	0.068	0.033

Note: Panel A estimates the first line of Equation (5) among pairs that carry another native- or stablecoin-intermediated route in the stated window. Panel B estimates the second line on all entrant pairs. The entry day is excluded from both follow-up windows. Controls are the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. Columns 3 and 6 weight pairs by route activity in the respective follow-up window; all other columns give each pair equal weight. Coefficients report the outcome change in percentage points (pp) associated with a 10 pp higher entry-day stablecoin share. Standard errors in parentheses use CR1 clustering by entry date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

4 How vehicle use becomes established

The pair decomposition locates the aggregate rotation in the pairs that become active and attract trading. The next question is how a vehicle becomes available inside those pairs and why its use persists. Longer routes carry the same rotation. The transition from Uniswap v1 to v2 separates inherited route structure from voluntary vehicle choice. Realised vehicle use then links the aggregate result to the capital on the two legs required to complete the route.

4.1 The rotation spans short and long routes

Figure 1 includes every complete route, with one observation for each intermediary position. Stablecoins gain across both short and long routes and within single- and cross-exchange routes. Appendix Table 18 reports the mutually exclusive vehicle choice on exact two-leg routes; Appendix Table 22, panel A, shows that stablecoin participation rises from 17.6% in 2024 to 46.0% in 2026 H1 when routes of every length enter. The full route network therefore moves in the same direction as the one-intermediary choice sample.

Panel B compares realised use with betweenness in the observed trading graph. WETH remains the highest-betweenness node in every annual graph even as its intermediary-position share falls and the combined USDC–USDT share rises. Binary network position therefore changes much less than realised intermediary use. An edge records that at least one leg connects two assets; local depth and execution cost determine how useful that connection is for a particular route.

4.2 Protocol architecture and inherited vehicle use

Uniswap v1 created pools between an ERC-20 token and ETH, so token-to-token trades crossed ETH. Uniswap v2 allowed arbitrary ERC-20–ERC-20 pools and made direct token-to-token trading possible.⁵ We reconstruct 217,003 connected v1 token-to-token routes, as reported in Appendix Table 8, panel A. In a March 2020 transaction, a trader selling 250 MKR received 439.13 ETH at one exchange contract and spent the identical ETH amount at a second for 49,892.40 DAI.⁶

Native-asset pairing remained pervasive after v2 opened direct token-to-token pools. In 2026, 95.5% of single-leg Uniswap v2 trades execute in WETH pools and 97.9% of newly traded token combinations include WETH. Appendix Table 8, panel B, reports both estimates. Native pairing therefore persisted after any two ERC-20 tokens could form a pool. We treat the v1–v2 transition as descriptive because v1 network thinning and migration to v2 changed available token combinations when the mandate disappeared. The pair-entry and bridge-depth estimates below examine settings in which stablecoin and WETH routes coexist.

4.3 Local bridge depth and route allocation

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). Here an indirect route needs two legs: one connecting the source to the vehicle and another connecting the vehicle to the destination. Capital on only one leg is insufficient. The weaker leg therefore determines the deposited-capital depth of the bridge. Deposited capital remains an equilibrium stock; a size-specific pretrade quote gives the closer measure of executable liquidity. Concentrated-liquidity providers can also change executable depth by moving capital along the price curve (Adams et al., 2021; Lehar and Parlour, 2025).

We date the first persistent stablecoin bridge for a pair that previously routed through WETH and had no observed stablecoin route. A bridge begins when DAI, USDC, or USDT has positive prior-calendar deposited capital on both required legs for at least 24 of the next 30 days. Panel A of Table 4 compares the 30 days before support with the first month and with days 30 through 119. Across 865 events, stablecoin route share rises by +7.93 pp in the first month and supported-value share by +3.11 pp. The corresponding count-share increase over days 30 through 119 is 8.68 pp.

The depth comparison follows each bridge after support appears. Let $K_{av,t-1}$ denote prior-calendar deposited capital on the leg joining endpoint a to vehicle v . The stablecoin set is $\mathcal{S} = \{\text{DAI}, \text{USDC}, \text{USDT}\}$. For pair (i, j) , define

$$\begin{aligned} D_{et}^S &= \max_{v \in \mathcal{S}} \min\{K_{iv,t-1}, K_{vj,t-1}\}, & D_{et}^W &= \min\{K_{iW,t-1}, K_{Wj,t-1}\}, \\ Q_{et} &= \frac{D_{et}^S}{D_{et}^S + D_{et}^W}, & S_{et}^R &= \beta Q_{et} + \alpha_e + \lambda_{m(t)} + \eta_{a(t)} + \varepsilon_{et}. \end{aligned} \tag{6}$$

⁵See Adams, Zinsmeister, and Robinson, *Uniswap v2 Core*, pp. 1–2.

⁶Transaction [0x4dca160d...96a0ca16](#), block 9,674,728, March 15, 2020. The transaction is the largest ETH route among the 129,810 mandate-era token-to-token transactions whose two ETH legs match exactly. The exact exchange registry identifies the endpoint tokens; public transfer records provide an independent cross-check.

Here W denotes WETH, S_{et}^R is stablecoin route share, $m(t)$ is calendar month, and $a(t)$ groups days since support into seven-day intervals. Event effects compare changes in relative depth within the same pair and route scope.

Panel B of Table 4 reports the relation. During the first month, the best stablecoin bridge has less than one-tenth of WETH depth on 7,752 of 11,327 comparable pair-days. Stablecoins carry 2.4% of routes in that range. Their route share reaches 53.0% when stablecoin depth matches WETH depth and 69.9% when it reaches twice WETH depth. Within the same bridge event, a 1 pp increase in Q_{et} is associated with +0.64 pp higher stablecoin route share.

Panel C separates support from use. Only 8.3% of the 865 events use one of their supported stablecoins on the support date; 52.8% do so within 30 days and 60.1% within 120 days. First-month use is 42.6% when stablecoin depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. Controls for prior WETH activity, endpoint direction, route scope, the supported stablecoins, and event month leave a +25.2 pp difference. Among pairs with two non-stablecoin endpoints, the difference is +35.8 pp.

Figure 2 follows the 246 events whose supported stablecoin is first used 7 to 119 days after persistent support. Stablecoin bottleneck capital rises +0.86 log points during the week before use and falls -0.44 over the following week. Relative to the matched WETH bridge, the corresponding changes are +0.78 and -0.46 ; the pre-minus-post contrast is +1.24. Appendix Table 25 locates the pre-use increase: 92.5% of stablecoin weak-leg capital on the first-use day lies in pools active one week earlier, while 7.5% lies in newly active pools. The pattern is consistent with providers deepening a bridge in anticipation of route demand, traders waiting for usable depth, or both forces operating together. Most formation occurs through deeper existing pools, with a smaller pool-activation margin. Because first use is jointly determined with capital, we treat the timing as suggestive evidence about formation and leave the two forces observationally combined.

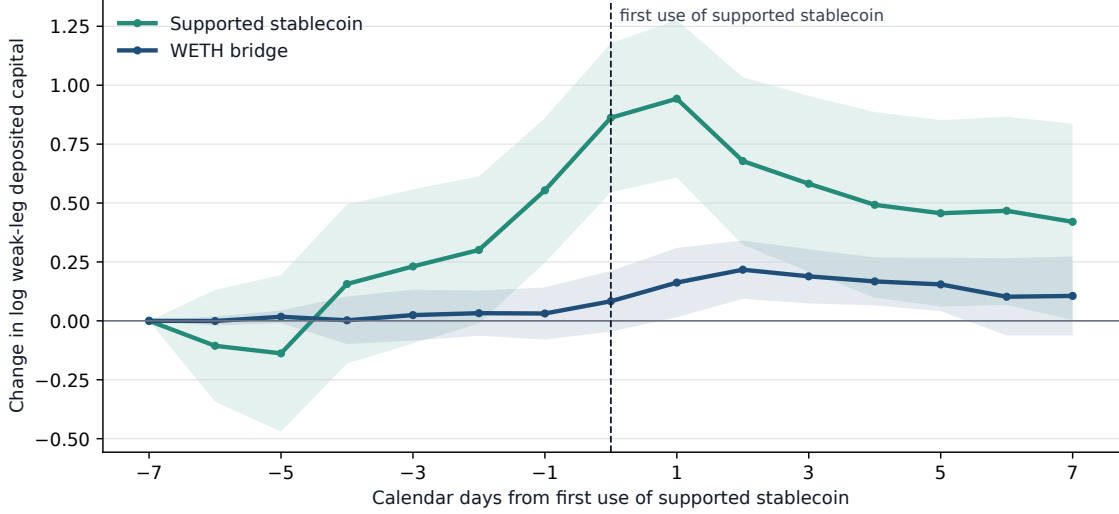
These results connect pair formation to local liquidity. A stablecoin route often becomes technically available while its bridge remains shallow. Use rises sharply as the two-leg bridge approaches WETH depth, and capital builds before first observed use. The post-entry persistence estimates remain similar among pairs with more entry-day trades (Appendix Table 23). At later horizons, route use and bridge depth remain positively related in both chronological and time-reversed comparisons (Appendix Table 26), consistent with persistent equilibrium comovement. The next section places current exact prices and lagged full-range constant-product capital in the same vehicle-choice regression.

Table 4: Stable-bridge establishment, relative depth, and route allocation

Outcome	First 30 days	Days 30–119
<i>Panel A. Changes around first persistent stable support</i>		
	+7.93***	+8.68***
Stable route share [pp]	(1.88)	(2.57)
	+3.11**	+6.94
Stable supported-value share [pp]	(1.53)	(4.53)
	−0.20***	−0.39***
log(1 + native routes per active day)	(0.04)	(0.06)
	−0.10**	−0.29***
log(1 + total routes per active day)	(0.04)	(0.05)
	−0.17	−0.61***
log(1 + native supported value per active day)	(0.11)	(0.19)
	+0.06	−0.32*
log(1 + total supported value per active day)	(0.10)	(0.18)
Bridge events	865	818
<i>Panel B. Stable-bridge competitiveness relative to WETH</i>		
	+0.64***	+0.75***
Relative-depth slope [pp per +1 pp]	(0.07)	(0.05)
Stable route share, depth < 0.1 × WETH [%]	2.4	2.4
	53.0	61.2
Stable route share, depth ≥ WETH [%]	(4.6)	(13.2)
	69.9	70.7
Stable route share, depth ≥ 2 × WETH [%]	(2.3)	(9.3)
Active pair-days	11,327	17,027
<i>Panel C. Adoption of supported stablecoin after persistent support</i>		
Outcome	First 30 days	First 120 days
Supported stablecoin used, depth < 0.1 × WETH [%]	42.6	50.5
Supported stablecoin used, depth ≥ 0.1 × WETH [%]	84.1	89.7
	+41.55***	+39.17***
Difference [pp]	(4.95)	(4.48)
	+25.16***	+23.47***
Difference with controls [pp]	(5.76)	(5.37)
	+35.76***	+40.26***
Difference, no stablecoin endpoint [pp]	(11.89)	(11.66)

Note: The unit is a pair by route scope around the first persistent DAI, USDC, or USDT bridge. Support requires positive prior-calendar deposited capital on both legs in Uniswap v2 or SushiSwap v2 on at least 24 of the next 30 calendar days. Every pair has earlier WETH-mediated routes and no earlier observed stablecoin route. Panel A reports changes from the prior 30 days. Panel B compares the best stablecoin weak-leg capital with WETH weak-leg capital and reports Equation (6). Panel C records use of a stablecoin in the event’s support set within 30 or 120 days. Standard errors are two-way clustered by pair and date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock.

Figure 2: Bridge capital accumulates before first use of the supported stablecoin



Note: The sample contains 246 pair-by-route-scope events whose first use of a stablecoin in the event’s support set occurs 7 to 119 calendar days after persistent support. Each line is the equal-event mean change from day -7 in $\log(1 + \text{deposited capital in USD})$ on the weaker leg. Stablecoin depth is the largest bottleneck among the supported stablecoins; WETH depth is the matched incumbent bottleneck. Capital is lagged to the prior calendar day. Shading gives 95% confidence intervals from standard errors two-way clustered by pair and adoption date.

5 Exact prices, full-range capital, and vehicle retention

Executable price differences can persist when trading is segmented across cryptocurrency markets (Makarov and Schoar, 2020). We therefore compare stablecoin and WETH routes at the same input and pretrade state. For each observed two-leg route on the fifteenth day of every month from June 2020 through June 2026, we quote the best WETH route and the best stablecoin route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. The observed quote must reproduce realised output within one basis point; both vehicle families must be feasible; every observed and hypothetical leg must remain below 5% own-price impact; leg values must agree within 20%; and input value must be at least \$100. Appendix B gives the pricing equations, Appendix D calibrates the common support rule, and Appendix Table 13 reports the broader price frontier.

We call a family the incumbent when it alone carries the pair on the first observed vehicle day and begin measuring retention 30 days later. For route r , pair p , and date t , the joint price-and-capital specification is

$$R_{rpt} = \beta_O A_{rpt}^O + \beta_K A_{pt-1}^K + \beta_{OK} A_{rpt}^O A_{pt-1}^K + \beta_q \log q_r + \alpha_p + \delta_t + \varepsilon_{rpt}. \quad (7)$$

Here R_{rpt} equals one when the route retains the incumbent family. The exact-output advantage is $A_{rpt}^O = [10^4 (O_{rpt}^{\text{inc}} - O_{rpt}^{\text{chall}}) / O_{rpt}^{\text{chall}}]_{-1000}^{1000} / 100$, so one unit represents 100 basis points. For family v , weak-leg capital is $K_{pt-1}^v = \min\{K_{i,v,t-1}, K_{v,o,t-1}\}$, where each leg aggregates prior-calendar

deposited capital across full-range Uniswap v2 and SushiSwap v2 pools. Its relative advantage is $A_{pt-1}^K = \{K_{pt-1}^{\text{inc}} / (K_{pt-1}^{\text{inc}} + K_{pt-1}^{\text{chall}}) - 1/2\} / 0.10$, so one unit represents a 10-percentage-point shift from an equal capital split. Pair fixed effects α_p absorb persistent pair-level routing preferences, and date fixed effects δ_t absorb market-wide conditions.

Table 5 reports the route-choice estimates. Column 1 relates contemporaneous price leadership to route choice; the coefficient on stablecoin price leadership is +57.59 pp. Among mature pairs with an exclusive incumbent, the incumbent carries 93.3% of routes when it leads on exact output and 27.2% when the challenger leads. In column 2, the within-pair, within-date coefficient on challenger leadership is -58.08 pp. Column 3 estimates Equation (7). With price and capital in the same regression, the coefficient on a 100-basis-point incumbent output advantage is +10.13 pp, and the coefficient on a 10-percentage-point incumbent lagged full-range constant-product capital advantage is +2.77 pp. Column 4 adds their interaction. Its coefficient is -0.44 pp (standard error 0.26 pp), consistent with lagged capital carrying a stronger retention association when the incumbent’s current quote is weaker.

The same comparison gives the output consequence of vehicle choice. Panel A of Table 6 shows that 12.9% of routes use the family that returns at least one basis point less. Conditional on that choice, the median shortfall is 27.2 bp and the 90th percentile is 171.5 bp; weighting every contestable route by input value gives 7.4 bp. Among routes that retain a mature exclusive incumbent, 10.7% leave more than one basis point of exact output, with a 17.1 bp conditional median and 3.3 bp across all retained routes (panel B). Panel C reports value-weighted shortfalls of 16.8 bp for pairs aged 0–89 days, 19.0 bp for pairs aged 90–364 days, and 2.1 bp after one year. The price cost is concentrated earlier in a pair’s life. Established routes remain responsive to public prices, while weak-leg capital retains an independent association with vehicle use.

Appendix Table 13 widens the same-state opportunity set beyond the two named vehicle families. Panel A first opens additional venues for the observed vehicle and then opens every named public vehicle plus the direct route. The final expansion adds only 2.0 pp to the share of routes with a gross-output improvement, while the median improvement across all better routes is 21.9 bp. Panel B shows a route-weighted stablecoin-share change of 1.2 pp. The broader comparison changes the venue classification more often than the intermediary asset.

Receipt gas adds a fixed toll to the extra leg and can affect both trading costs and liquidity concentration (Caparros et al., 2024). The realised toll becomes smaller relative to observed stable-vehicle trades over the sample; Appendix E reports its gas-unit, dollar, and basis-point magnitudes. Hypothetical route gas is unavailable; the same-state price comparison remains gross of gas.

The benchmark represents public on-chain liquidity in the declared exchange set. Private order flow, financing costs, losses from reverted transactions, and pools outside that set can alter a router’s choice. Across sampled Curve pool-days, pools outside the StableSwap model carry 34.2% of assessed volume, including 65.2% of native-leg volume and 21.1% of stablecoin-leg volume (Appendix Table 16, panel A). Hook-bearing or dynamic-fee Uniswap v4 pools, SushiSwap v3, and several Balancer families also use pricing states outside the benchmark. Appendix E reports the

Table 5: Exact prices, full-range weak-leg capital, and vehicle retention

Outcome; estimates [pp]	(1) Stablecoin chosen	(2) Incumbent retained	(3) Incumbent retained	(4) Incumbent retained
Stablecoin route has higher exact output	+57.59*** (2.84)			
Challenger route has higher exact output		-58.08*** (2.82)		
Challenger output lead $\times$ stable incumbent		+4.47 (7.82)		
Incumbent exact-output advantage [100 bp]			+10.13*** (0.77)	+11.35*** (1.15)
Incumbent lagged full-range capital-share advantage [10 pp]			+2.77*** (0.61)	+2.67*** (0.60)
Output advantage $\times$ capital-share advantage [100 bp $\times$ 10 pp]				-0.44* (0.26)
Log input value [USD]	+0.32 (0.43)	-0.66 (0.46)	-1.32* (0.78)	-1.10 (0.78)
Dependent mean [%]	26.5	73.7	69.7	69.7
Within R^2	0.372	0.382	0.153	0.156
Observations	49,212	34,439	17,778	17,778
Pair clusters	2,596	1,981	678	678
Date clusters	73	73	73	73
Pair fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date
Exclusive entry, age $\geq$ 30 days	No	Yes	Yes	Yes
Prior-day full-range capital positive, both vehicles	No	No	Yes	Yes
Minimum absolute output difference [bp]	1	1	1	1
Continuous output gap cap [bp]			1,000	1,000

Note: The unit of observation is a realised two-leg route on the fifteenth day of each month from June 2020 through June 2026. For the same ordered endpoint pair, input amount, and pre-transaction state, we quote the best WETH route and the best stablecoin route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Input value is at least \$100, leg values agree within 20%, both vehicle families are feasible, and every observed and quoted leg remains below 5% own-price impact. Ties within one basis point are omitted, and all prices are gross of gas. Column 1 is a linear probability regression of stablecoin use on an indicator that the stablecoin route returns more exact output. Columns 2 through 4 study retention after a pair first uses one vehicle family exclusively and that first use is at least 30 days old. Column 2 includes an indicator that the challenger returns more exact output and its interaction with a stablecoin incumbent. Columns 3 and 4 use observations with positive prior-day full-range constant-product weak-leg capital for both families; column 3 estimates Equation (7), and column 4 adds the price-capital interaction. Weak-leg capital is the smaller deposited-capital stock across the two required legs, aggregated across Uniswap v2 and SushiSwap v2; the capital-share regressor measures the incumbent’s deviation from one-half. The exact-output advantage is measured per 100 basis points and capped at plus or minus 1,000 basis points; the capital-share advantage is measured per 10 percentage points; input value enters in logs. Every column includes ordered endpoint-pair and date fixed effects. Standard errors in parentheses are clustered by pair and date. Coefficients and standard errors are in percentage points. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: Exact-output shortfalls across feasible vehicle families

Measure	Estimate	Routes
<i>Panel A. All contestable routes</i>		
Observed family has an exact-output shortfall above 1 bp [%]	12.9%	52,477
Conditional median shortfall [bp]	27.2	6,786
Conditional 90th-percentile shortfall [bp]	171.5	6,786
Input-value-weighted mean shortfall [bp]	7.4	52,477
<i>Panel B. Routes retaining a mature exclusive incumbent</i>		
Observed family has an exact-output shortfall above 1 bp [%]	10.7%	27,215
Conditional median shortfall [bp]	17.1	2,901
Input-value-weighted mean shortfall [bp]	3.3	27,215
<i>Panel C. Input-value-weighted mean shortfall by pair age</i>		
Pairs aged 0–89 days [bp]	16.8	13,159
Pairs aged 90–364 days [bp]	19.0	13,166
Pairs aged at least 365 days [bp]	2.1	26,152

Note: The unit is a realised two-leg route in the common-support sample underlying Table 5. For the same ordered endpoint pair, input amount, and pre-transaction state, we compare the best route through WETH with the best route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Every observed and quoted leg remains below 5% own-price impact, and input value is at least \$100. A shortfall occurs when the observed vehicle family returns more than one basis point less exact output than its feasible rival. Conditional quantiles use routes with such a shortfall; input-value-weighted means use every route in the stated sample and assign zero below the one-basis-point threshold. Panel B retains routes at least 30 days after a pair first uses one vehicle family exclusively and on which the route keeps that incumbent family. Panel C measures age from the pair’s first observed supported date. Estimates are gross of gas and exclude private order flow and venues outside the quoted set.

remaining coverage limits.

6 Discussion and policy implications

We group stablecoins by their shared vehicle function while retaining the token that supplies each route. USDC, USDT, DAI, and other stablecoins can target the same unit of account while differing in issuer, backing, redemption, peg risk, and local pool liquidity. The family-level result measures the rise of tokenised-dollar intermediation; the issuer-level estimates show how that role is allocated within the family. Stable-to-stable endpoint pairs identify this distinction directly because USDT supplies most of their increase in routed value.

A one-for-one peg aligns prices while preserving currency identity. The Bahamian dollar, for example, is conventionally pegged at one Bahamian dollar per U.S. dollar and remains a separately issued currency with its own monetary and exchange-control arrangements ([International Monetary Fund, 2025](#)). The same distinction disciplines our stablecoin aggregation. A currency-equivalent measure would consolidate dollar-pegged instruments by unit of account; our route measure records competition among separately issued dollar instruments and between that family and WETH.

This distinction may also affect liquidity supply. While two stablecoins hold their pegs, their relative price typically varies less than either token’s price against WETH. In a constant-product pool, smaller relative-price movements imply smaller loss versus holding, commonly called impermanent or divergence loss. That risk profile may make stablecoin–stablecoin links attractive to liquidity providers and help tokenised dollars build dense cross-issuer markets. Lower prior endpoint–vehicle risk predicts deeper constant-product bridge capital at the measurement date and 30 days later (Appendix Table 27, panel A, columns 1–4). Stablecoin bridges nevertheless have higher median relative volatility than WETH bridges and carry the lower risk in fewer than one third of comparable pair-months (panel B). Divergence risk helps explain where capital sits; it does not account for the aggregate stablecoin rotation. Impermanent loss remains distinct from loss-versus-rebalancing, the running adverse-selection cost of stale AMM prices ([Milionis et al., 2022](#)).

A newly issued stablecoin also gains convertibility by linking to incumbent dollar tokens, so issuer-sponsored or affiliated-market-maker capital may seed cross-issuer pools alongside independent liquidity supply. Deposited capital reveals the pool stock; provider identities, funding relationships, entry prices, fee income, hedges, and expectations determine profitability. We therefore use these channels as economic interpretations of pool formation.

Vehicle use connects otherwise unrelated pairs to the liquidity and issuer conditions of the asset in the middle. This concentration can lower trading costs through deeper two-leg markets while transmitting a disruption in those markets across many endpoint pairs. The pair-entry estimates show why route allocation may adjust slowly after those links form. These are market-structure implications of observed route allocation. Welfare effects require additional evidence on user surplus, provider returns, and the incidence of stablecoin risk.

Payment-system design faces the same extensive margin. Nexus assigns foreign-exchange conversion to a provider, and Rialto uses a third currency when direct exchange is unavailable ([Bank for International Settlements Innovation Hub, 2024, 2025](#)). The route evidence suggests that access policy affects more than the price quoted inside an existing relationship. It also affects which

two-sided markets form early enough and deeply enough to become the bridge for new relationships.

7 Conclusion

Stablecoin vehicle dominance rises sharply on decentralised exchanges. Between matched January–June periods in 2024 and 2026, the equal-day stablecoin share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value (Appendix Table 18). In the pooled decomposition, movements toward and away from stablecoins offset inside continuing pairs, while activity shifting across continuing pairs contributes +8.4 pp by route count. Pairs first observed after 2024 H1 contribute +20.1 pp before reactivation, vehicle-role turnover, and exit reduce the period-specific net to +17.8 pp (Table 2, panel A). First-observed pair entry contributes +21.9 pp by routed value (panel B). Longer routes show the same movement.

The vehicle used at pair entry predicts later use after the entry day is removed from every outcome (Table 3). First-month stablecoin use rises from 42.6% below one-tenth of WETH depth to 84.1% once that threshold is reached (Table 4, panel C); capital on the weaker leg builds before first use (Figure 2). The exact stablecoin-versus-WETH contest sharpens the persistence result. The incumbent-retention coefficient on challenger price leadership is -58.08 pp, while the coefficient on a 10 pp incumbent lagged full-range constant-product capital advantage is +2.77 pp after current output advantage is held fixed (Table 5, columns 2 and 3). The price–capital interaction is -0.44 pp in column 4 and points toward a larger capital association when the incumbent’s quote is weaker. Among routes that retain a mature incumbent, the input-value-weighted exact-output shortfall is 3.3 bp (Table 6, panel B). Incumbent retention tracks the current price ranking and remains higher where the incumbent has more lagged full-range constant-product capital.

Aggregate dominance combines use within continuing relationships with the vehicle identity of new ones. The large stablecoin rotation comes mainly from pair entry and where trading grows across continuing pairs. Early vehicle use and two-sided depth tie new relationships to the markets around them; contemporaneous prices determine how firmly that tie holds. Currency competition therefore operates inside established relationships and when new relationships form the markets that later support them.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol’s factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, pool-swap legs are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one exchange between its pair endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. Pool identities enter reconstruction; the unified route panel retains the ordered venue and token sequence together with the pair. It does not infer that sequence from the transaction sender or router label.

Disconnected event groups become more common late in the sample. They comprise 0.8% of reconstructed components in 2024 and 6.2% in 2026. In 2026 the rate is 9.4% among components touching Uniswap v4 and 4.7% among other components. Table 7 uses the internally connected token flow of each group as a separate route. This alternative preserves the principal economic result: the 2026 stablecoin share is 43.7% by count and 77.3% by value, both slightly above the connected-transaction estimates.

Table 7: Disconnected transaction components and vehicle dominance

Sample boundary or estimate	2024	2025	2026
<i>Panel A. Share of reconstructed components excluded</i>			
All components	0.8%	3.7%	6.2%
Components touching Uniswap v4		11.6%	9.4%
Other components		2.7%	4.7%
<i>Panel B. Stablecoin share among native and stable two-leg vehicles</i>			
	2024	2026	Change [pp] (s.e.)
Connected routes, count	16.9%	42.3%	+25.4 (1.05)
Connected routes, value (20% agreement)	32.7%	76.5%	+43.9 (2.02)
Each valid component, count	16.9%	43.7%	+26.8 (1.06)
Each valid component, value (20% agreement)	32.9%	77.3%	+44.4 (1.93)

Note: Panel A reports the share of reconstructed components belonging to transactions with more than one disconnected token-flow group. Uniswap v4 begins during 2025, so its 2024 cell is empty. Panel B compares the principal connected-transaction rule with an alternative that treats each internally well-formed component as a separate route. Stablecoin shares use native and stable two-leg vehicles as the denominator and equal-weight matched January through June dates in 2024 and 2026. Value estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth. The alternative assigns no unobserved handoff between components.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. A failed valuation removes the observation from the value calculation while preserving the observed route. Appendix F gives the separate exclusion for self-returning routes.

A hypothetical AAVE–WETH–USDC–UNI route makes the units concrete. Suppose one AAVE is worth \$100, one WETH \$2,500, one USDC \$1, and one UNI \$10. If one AAVE returns 0.04 WETH, the second leg returns 100 USDC, and the third returns 10 UNI, then $q = \$100$, $Q_r = 10$ UNI, and $O_r = \$100$. Each leg carries \$100 on both sides. The route value ν_r is also \$100 because it averages the \$100 source and destination values. In the data, the same calculation uses transaction-time prices and allows limited measurement error under the value-agreement rule.

Historical token precision cannot be verified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

Table 8: Protocol architecture and native-asset pairing

Measure	Estimate
<i>Panel A. Uniswap v1: ETH intermediation required</i>	
Reconstructed token-to-token routes [count]	217,003
<i>Panel B. Uniswap v2: arbitrary token pools allowed</i>	
Single-leg trades executed in WETH pools, 2026 [%]	95.5
Token combinations first traded with WETH, 2026 [%]	97.9

Note: Panel A reports connected Uniswap v1 token-to-token routes reconstructed from the exchange registry and matched ETH legs. The protocol allowed only ETH–token pools, so each such route crossed ETH. Panel B reports Uniswap v2 activity during January through June 2026. A single-leg trade uses a WETH pool when either pool token is WETH. A token combination first traded with WETH when either token in the newly active combination is WETH. V2 allowed arbitrary ERC-20 token pools throughout this period.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool’s state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve’s amplification coefficient and Balancer’s weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the

transaction.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p 's input- and output-token reserves immediately before transaction e , Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \quad \varphi_p = \frac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.⁷

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated routes at the same state and dollar input notional,

$$10^4 \Delta C_{i,o,k,q,e}^D = 10^4 \frac{O_{i,o,q,e}^D - O_{i,o,k,q,e}^I}{O_{i,o,q,e}^D}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\tilde{R}_{i,p,e^-}$ and $\tilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\tilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \quad \tilde{R}_{o,p,e^-} = L_{p,e^-} \sqrt{\rho_{p,e^-}}.$$

⁷The SushiSwap v2 pool contract implements the same adjustment; see [UniswapV2Pair.sol](#).

Table 9: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine token combinations, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \tilde{R}_{o,p,e^-}}{\tilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

until the trade reaches the next initialized tick. Range endpoints lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \quad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract’s integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrtPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 9 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.⁸ The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing row of Table 9 reproduces realised output within 0.001% in the Maximum column.

⁸See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.⁹ For n tokens with normalized pre-transaction balances R_{j,p,e^-} , amplification A_p , and invariant D_p ,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \rightarrow 0$ recovers the constant product and $A_p \rightarrow \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i . Defining $D_\Pi = D_p \prod_j D_p / (n R_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + n D_\Pi) D_p}{(A_p n - 1) D_p + (n + 1) D_\Pi}, \quad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2 R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is off by a factor of 10^{12} . Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1) \varphi_p,$$

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it. We identify it from data because every other quantity in the invariant is observed and every Curve swap reports both amounts,

$$\hat{A}_p = \arg \min_{A_p} Q_{0.90} \left\{ \frac{|Q_p(i \rightarrow o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o} \right\},$$

fitted on the first half of a pool-day's trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this component exercise.

Table 10 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

⁹See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

Table 10: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\hat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\hat{A}_p$ is reported in the module’s convention, which carries the on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,¹⁰

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \quad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pool with the same reserves therefore return the same quote. Balancer’s weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.¹¹ The counterfactual design applies the common 5% own-price-impact cutoff in Equation (9) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool’s owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which appears directly in the preceding validation: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p \Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both as unknowns to be recovered from the trades a pool actually executed, the move Curve’s amplification coefficient needed in Section B.3, and we keep a reported value only where it survives that test. Reconstructing a pool’s state from its own event history follows [Lehar and Parlour \(2025\)](#); what the weighted family adds is that two arguments of the pricing function

¹⁰See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

¹¹See the input-ratio bound in Balancer’s [WeightedMath.sol](#).

are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day's trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 11 is measured on the evaluation set. For a trial parameter pair ϑ ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90} \left\{ \frac{|Q_p(i \rightarrow o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p \right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. The upper tail does the gating for the reason it does at Curve, and the coverage condition stops a parameter that prices one corner of a pool's activity from standing for the pool.

Each token combination's weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\hat{\theta}_{ab,p} = \arg \min_{\theta} \mathcal{E}_p(\theta \text{ on } \mathcal{O}_p^{a \rightarrow b}, \theta^{-1} \text{ on } \mathcal{O}_p^{b \rightarrow a}),$$

searched over a logarithmic grid spanning Balancer's own weight bounds and refined by ternary search in the log exponent, and attempted only for token combinations carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as an exact fit. The fee $\hat{f}_p$ is recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its trades on reported parameters is never handed a fitted one,

$$\hat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \leq \bar{e}, \\ (\theta_p, \hat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \hat{f}_p) \leq \bar{e}, \\ (\hat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\hat{\theta}_{\cdot,p}, f_p) \leq \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \quad \bar{e} = 0.1\%. \quad (8)$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the token combinations that happened to fit and then quoted on the rest.

The frequency with which the parameter search binds measures the sufficiency of reported pool state. Of the 256 pool-days accepted across the twelve validation days, 209 satisfy the error threshold on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-token-combination weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

Table 11: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day’s event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day’s opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

We set $\bar{e}$ at 0.1% instead of the 1% Curve uses because the achieved errors separate cleanly there. The threshold separates weighted pools, whose held-out errors range from 10^{-9} to 0.004%, from stable, composable-stable, and elliptic-curve pools, whose held-out errors range from 0.05% to 1.08% and have a 99th percentile of 12%. At a 1% threshold, fitted weight ratios would admit the second group. The 0.1% rule excludes it. Curve’s failed validation arose from an over-permissive parameter range; here, an over-permissive error threshold would produce the same mistake.

Across the twelve date rows of Table 11, the Backward column returns a median error of 0.0000% and the Pools priced / excluded column identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. The exercise therefore validates the model for identified weighted pools, not for Balancer as a whole.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. The exercises below validate that procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p ’s stored period-end reserve vector and $\Delta_{p,e}$ the vector of net signed token

amounts in event e . The state before event e is the stored value with every later event removed,

$$\mathbf{R}_{p,e}^{\text{pre}} = \mathbf{R}_p^{\text{end}} - \sum_{e' \geq e} \Delta_{p,e'} = \mathbf{R}_{p,e+1}^{\text{pre}} - \Delta_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period's stored value as the current period's opening state and replays forward,

$$\tilde{\mathbf{R}}_{p,e}^{\text{pre}} = \mathbf{R}_{p,\text{prev}}^{\text{end}} + \sum_{e' < e} \Delta_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances; the ordered replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}_{p,1}^{\text{pre}} - \mathbf{R}_{p,\text{prev}}^{\text{end}} \right) \oslash \mathbf{R}_{p,\text{prev}}^{\text{end}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 11 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Component-validation samples contain executed swaps and therefore pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. Let $\tilde{Q}_\ell$ denote leg ℓ 's output at the pool's marginal rate. We include a hypothetical leg only when

$$\pi_\ell(\Delta_i, e^-) = \frac{\tilde{Q}_\ell(\Delta_i, e^-) - Q_\ell(\Delta_i, e^-)}{\tilde{Q}_\ell(\Delta_i, e^-)} \leq \bar{\pi}, \quad \bar{\pi} = 0.05. \quad (9)$$

Table 12: Size relative to depth in the validation population

Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quotes were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

The ceiling is calculated under the pricing rule of the pool family the leg would traverse; contract-specific limits add further restrictions.

For constant-product swaps, input relative to reserves provides a transparent size–depth comparison,

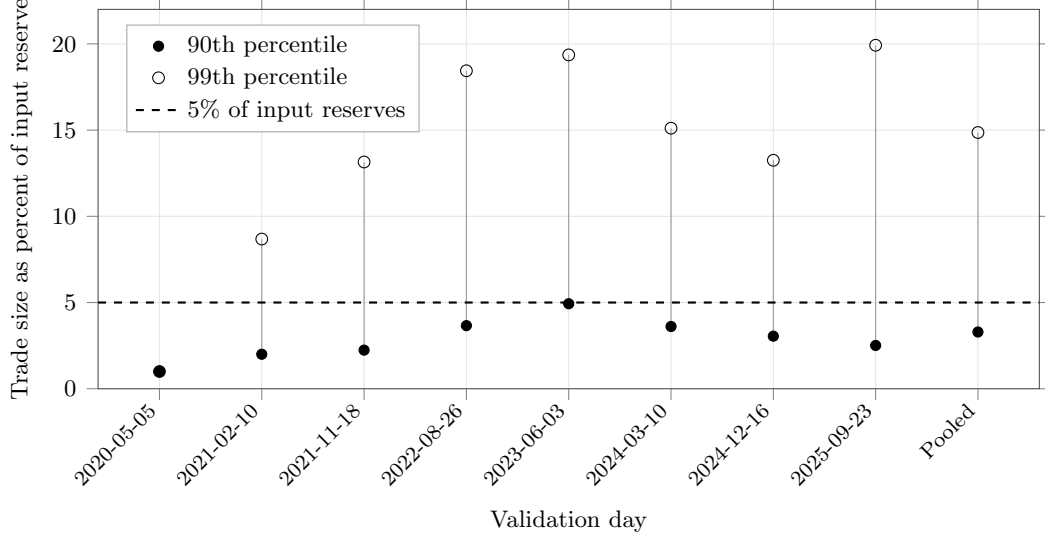
$$\iota_{i,p,e^-} = \frac{\Delta_i}{R_{i,p,e^-}},$$

the input amount as a fraction of the input-side reserve. Table 12 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

In the Pooled row of Table 12, five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86%; the same cutoff lies between those columns on seven of the eight date rows. Figure 3 shows the daily distributions. This calculation locates the cutoff in the empirical size distribution. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family’s pricing function.

Equation (9) applies the same 5% own-price-impact ceiling to every family. The component exercises establish pricing accuracy for realised swaps within their stated samples; they do not measure quote error on unexecuted routes.

Figure 3: Constant-product trade size relative to reserves



Note: The statistic is a leg’s input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Realised routes identify the assets that carried each trade. The price frontier holds the source i , destination o , dollar input q , and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O_{i,o,q,e}^D = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p(i \rightarrow o, \Delta_i(q, e); \chi_{p,e^-}),$$

$$O_{i,o,k,q,e}^I = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'} \left(k \rightarrow o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p(i \rightarrow k, \Delta_i(q, e); \chi_{p,e^-}); \chi_{p',e^-} \right).$$

The vehicle-route shortfall relative to direct execution is

$$\Delta C_{i,o,k,q,e}^D = \frac{O_{i,o,q,e}^D - O_{i,o,k,q,e}^I}{O_{i,o,q,e}^D},$$

reported in basis points as $10^4 \Delta C_{i,o,k,q,e}^D$. The comparison extends the realised-versus-optimised allocation benchmark of [Xi and Moallemi \(2026\)](#) from parallel pools connecting the same assets to sequential routes through an intermediary. Swaps and liquidity changes are replayed in on-chain order so every quote uses information available before the trade.

The opportunity set expands in three stages: alternative paths using the same vehicle and venue families as the observed route, paths using the same vehicle across Uniswap v2, SushiSwap v2, and Uniswap v3, and paths using any named public vehicle or the direct connection across

Table 13: Exact pre-transaction prices and vehicle identity

<i>Panel A. Coverage and quote validation</i>				
Linear routes in the exact venue set [%]				93.4
Mapped observed routes reproduced within 1 bp [%]				89.2
<i>Panel B. Nested opportunity set</i>				
	Realised route	Same vehicle, used venues	Same vehicle, all exact venues	Any named vehicle or direct
Routes with more than 1 bp higher output [%]	0.0	6.6	44.5	46.4
Stablecoin vehicle share [%]	19.5	19.5	19.5	18.4
Full-set minus realised stablecoin share [pp]			Route weighted	−1.16*** (0.19)
			Input-value weighted	−2.09*** (0.31)
Routes / dates				777,203 / 73

Note: The sample contains exact two-leg routes executed on Uniswap v2, SushiSwap v2, or Uniswap v3 on the fifteenth day of each month from June 2020 through June 2026. Source, destination, input amount, and pre-transaction pool state remain fixed. Each observed quote reproduces realised output within one basis point. Leg values must agree within 20%, input must be at least \$100, and every observed or hypothetical leg must remain below 5% own-price impact. The public vehicle set contains WETH, USDC, USDT, DAI, and WBTC; a realised vehicle outside that set remains eligible. A route changes vehicle only when gross output improves by more than one basis point. Standard errors in parentheses are clustered by date. P-values use Holm adjustment across route- and input-value-weighted comparisons in five reported samples. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The three exact venues contain 93.4% of linear routes over the sampled dates, and 89.2% of mapped observed routes reproduce within one basis point. Coverage falls as other exchanges expand late in the sample. The comparison is gross of gas and excludes private order flow. 98 routes imply more than twice realised output under standard invariant pricing; removing them leaves every displayed share unchanged at one decimal place, and they are excluded from the median-gain calculation.

those exchanges. Each observed two-leg route must reproduce realised output within one basis point. Leg values must agree within 20%, input value must be at least \$100, and every observed or hypothetical leg must satisfy Equation (9).

Table 14 compares later route use with the independently quoted price leader.

E.1 Realised gas burden

Receipt gas measures the fixed execution toll attached to adding a vehicle leg. In the executed-route sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year’s median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, while the

Table 14: Current price leadership and entry-vehicle use

Vehicle at pair entry	Route weighted		Pair-day weighted	
	Entry vehicle leads	Other vehicle leads	Entry vehicle leads	Other vehicle leads
All pairs	93.4 [21,325]	22.4 [8,807]	94.6 [7,332]	24.5 [3,509]
Native	95.9 [19,207]	26.5 [6,750]	96.2 [6,451]	28.2 [2,738]
Stablecoin	70.7 [2,118]	9.0 [2,057]	82.7 [881]	11.6 [771]

Note: Each cell reports the percentage of observations that use the vehicle recorded at pair entry; brackets give the number of routes or pair-days. Observations are at least 120 days after entry. For the same pair, input amount, and pre-transaction state, we independently quote the best path through WETH and the best path through DAI, USDC, or USDT. Both paths must be feasible, every leg must remain below 5% own-price impact, input must be at least \$100, and the leading path must give more than one basis point of additional gross output. The comparison uses the fifteenth day of each month from June 2020 through June 2026 across exact Uniswap v2, SushiSwap v2, and Uniswap v3 states. It is gross of gas and excludes private order flow and other venues.

dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. At each stable-vehicle route’s own gas price, median stable-vehicle notional falls from \$832 to \$82, the median fixed toll falls from 44.3 bp to 16.2 bp, and the share of routes with a toll below 25 basis points rises from 39.5% to 55.7%. The extra leg adds gas units but becomes small relative to most observed stable-vehicle notionals. Because hypothetical route gas is unavailable, these estimates describe the realised fixed toll and leave the same-state price comparison gross of gas.

E.2 Exchange and pool-family coverage

Omitting an exchange can change the best direct route, the best intermediated route, or both. Panel A of Table 15 describes the volume recorded in nine local source series, including Uniswap v1 and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a \$456m day whose swaps sum to \$8, and a constant-product pool with \$255 of reserves reports \$1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 enters the full route panel but not the route-cost opportunity set because the price comparison requires a common pre-transaction state model. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. Panel B uses DeFiLlama’s daily protocol breakdown to compare the selected exchange families with total Ethereum DEX volume. Their combined share declines from 98.5% in 2020 to 77.5% in 2026 H1 and is 87.5% over the displayed period. This external comparison describes the families’ market reach; it does not establish transaction-level or pretrade-state coverage for every deployment.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose

Table 15: Exchange-source composition and Ethereum market coverage

<i>Panel A: shares within the nine retained source series</i>									
Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61
<i>Panel B: selected exchange families as a share of total Ethereum DEX volume</i>									
Period	2020	2021	2022	2023	2024	2025	2026 H1	Pooled	
Share (%)	98.5	92.2	92.3	86.6	87.2	82.4	77.5	87.5	

Note: Panel A reports percentage shares of volume recorded in nine local source series. Its weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Panel B sums the protocol families represented in the route panel and divides by total Ethereum DEX volume in DeFiLlama’s daily breakdown; the numerator includes Uniswap v1–v4, SushiSwap v2 and v3, Curve, Balancer v2, and Fluid. DeFiLlama’s protocol taxonomy need not match the exact transaction-level deployments in Panel A. The 2026 cell ends on June 30, and the pooled cell sums 2020 through 2026 H1. See [DeFiLlama’s DEX-volume series](#).

transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits 621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 pp. Table 16 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. In Panel B of Table 16, the excluded share of native-leg volume exceeds the stable-leg share by at least 33 pp in every sample year and by 54 pp in the 2023 row. Tricrypto pools pairing WBTC and WETH with a stablecoin account for \$818m, or 44.6%, of the \$1.83bn excluded, including seven of the twelve largest excluded pool-days. These CryptoSwap pools concentrate the missing coverage where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool’s root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only 4.1% of its volume occurs on token combinations unavailable on another included exchange, never more than \$37,609 on one

Table 16: Curve volume outside the StableSwap comparison

<i>Panel A: composition of the exclusion</i>					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	\$1.038bn	56.6%	15.7%	65.2%
Stable	119	\$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	\$0.045bn	2.4%	4.4%	22.2%
<i>Panel B: excluded share of each leg’s volume, by year</i>					
Year	Native leg		Stable leg	Other volatile	
2021	85.49%		46.91%	15.23%	
2022	68.82%		30.21%	50.77%	
2023	59.11%		5.51%	56.60%	
2024	48.24%		0.94%	14.38%	
2025	47.47%		13.96%	10.52%	
2026	60.92%		9.04%	20.08%	

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool tokens are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023 peak. Table 11 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool’s own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (8).

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}_{\{k_0=k_{m(r)}\}},$$

does not represent a conversion between distinct endpoint assets. We exclude these routes from the dominance measures. The excluded population can contain cyclic arbitrage and other self-returning activity; the endpoint rule does not classify every route as arbitrage or wash trading. Multi-trade arbitrage can be bundled atomically (Daian et al., 2020), but identifying trader intent would require additional evidence.

Table 17 reports the distribution across every day with at least 200 multi-leg routes.

At the median, round trips account for 12.8% of multi-leg routes by count and 21.0% by value. At the 95th percentile the shares are 23.0% and 51.1%, and the value-share maximum reaches 97.7% on 13 October 2023. The wider upper tail by value is consistent with large self-returning routes

Table 17: Round-trip share of multi-leg routing across all eligible days

Quantile across days	Share by count	Share by value
Minimum	0.00%	0.00%
25th percentile	9.66%	15.44%
Median	12.82%	20.96%
75th percentile	16.28%	27.99%
95th percentile	22.97%	51.07%
Maximum	33.09%	97.73%
2019 median	0.00%	0.00%
2020 median	12.85%	12.77%
2021 median	13.39%	23.95%
2022 median	14.17%	25.82%
2023 median	9.18%	17.07%
2024 median	10.38%	19.83%
2025 median	16.21%	30.47%
2026 median	21.21%	20.08%

Note: Round trips as a share of multi-leg routes. The sample contains all 2,449 eligible dates from 15 May 2019 through 30 June 2026 and 359,631,951 routes, of which 68,927,053 are multi-leg. Dates with fewer than 200 multi-leg routes are omitted.

carrying disproportionate notional. Round trips are excluded before the count- and value-weighted dominance measures are formed.

G Additional composition and network results

Appendix Table 21 separates endpoint composition from intermediary identity inside stable-to-stable routes.

The full route panel supports two complementary measures. Intermediary-position share allocates all observed intermediary work across currencies. Route-participation share records the fraction of routes containing a currency and can sum above one across currencies because a longer route may contain several intermediaries. Panel A of Table 22 reports both measures for the native and stablecoin families. Panel B compares realised use with each named currency’s position in the

Table 18: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change [pp] (s.e.)
Route count	16.9%	42.1%	+25.2 (1.04)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.8 (2.01)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable. Daily shares receive equal weight over matched January–June dates in 2024 and 2026. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

Table 19: Alternative accounting for the route-count rotation

Component	Estimate [pp]
Pairs entering or leaving the sample	+10.0
Pairs gaining or losing a vehicle route	−0.4
Market activity shifting across continuing pairs	+7.7
Change in how often continuing pairs use a vehicle	+6.9
Stablecoin share within continuing vehicle-using pairs	+1.3
Total route-count change	+25.5

Note: The table factors the pooled route-count change into pair support, vehicle-route support, activity reallocation, vehicle incidence, and stablecoin share within continuing vehicle-using pairs. It is an alternative accounting of the same total reported in Table 2, panel A; its components use a different mass and are not added to the main decomposition.

Table 20: Time-window and endpoint restrictions on the stablecoin rotation

Panel A. Adjacent January–June comparisons, route count

H1 comparison	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit
2019–2020	+7.3	+0.0	+0.0	+3.0	+4.2
2020–2021	+11.6	+0.5	+11.3	−6.4	+6.2
2021–2022	+10.3	+0.5	+1.8	−1.5	+9.5
2022–2023	−6.4	−0.4	−1.7	−4.5	+0.1
2023–2024	−5.0	−0.4	+1.6	−6.8	+0.5
2024–2025	+8.0	−0.1	+2.9	−2.1	+7.4
2025–2026	+17.4	+1.0	+5.0	−1.1	+12.4

Panel B. Neither endpoint is WETH or a stablecoin

	2024 H1	2026 H1	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit
Route count	1.1	9.0	+8.0	+0.2	+0.6	−0.0	+7.2
Supported value	0.9	39.1	+38.1	−0.3	+1.5	+3.1	+33.8

Note: Both panels apply the exact decomposition in Equation (3); endpoint-year shares are percentages, and all changes and components are in pp. Panel A uses equal-weight matched January–June dates in each adjacent pair of years. Panel B compares matched January–June dates in 2024 and 2026 after removing every pair with WETH or a stablecoin at either endpoint. Supported value requires source, intermediary, and destination values to agree within 20%. The across-pair component changes weights among continuing pairs, the support shift changes the aggregate weight on continuing pairs, and pair entry/exit is net of pairs present in only one endpoint year. Components use unrounded values and sum to the total change.

Table 21: Endpoint direction and stablecoin intermediary identity

	Route count [pp]	Routed value [pp]
<i>Panel A. Contribution by endpoint direction</i>		
All other pairs	+15.2	+20.6
One native, one stable endpoint	+9.2	+10.1
Two stable endpoints	+1.0	+13.2
Total stablecoin-share change	+25.4	+43.9
<i>Panel B. Stablecoin identity within two-stable-endpoint pairs</i>		
USDT	+0.6	+13.7
USDC, DAI, and other stablecoins	+0.4	-0.5
All stablecoin intermediaries	+1.0	+13.2
Top-decile-trimmed daily mean	–	+12.3
Pooled route-value weighting	–	+12.7

Note: Each entry is a contribution to the equal-day stablecoin-share change between matched January–June dates in 2024 and 2026. Panel A partitions the full native-or-stable intermediary sample by pair endpoints. Panel B partitions the stablecoin contribution within pairs whose two endpoints are stablecoins; its first two rows sum to the third. Routed value requires source, intermediary, and destination values to agree within 20%. The trimmed row removes the largest decile of daily contributions in each year. The pooled row weights routes by supported value across the matched dates. Components use unrounded values.

trading graph.

Table 22, panel A, shows the same stablecoin rotation under both all-route normalisations. Panel B shows a different movement in graph position. WETH remains the first-ranked betweenness node in every year, even as USDC and USDT gain realised intermediary use. The result separates the existence of leg connections from the depth and execution conditions that determine route allocation.

The issuer split in Appendix Table 21, panel B, keeps the token-level result beside the endpoint composition it explains. Together with Table 22, these exhibits show that the rotation spans longer routes and separately issued stablecoins even though WETH remains the graph’s most central connector.

Table 22: All-route participation and trading-network position

<i>Panel A. All complete route lengths</i>				
Year	Currency family	Intermediary-position share [%]	Route-participation share [%]	Routes using family
2020	Native	71.7	78.2	2,915,214
	Stablecoin	25.4	26.0	970,851
2024	Native	75.3	79.5	6,894,487
	Stablecoin	17.2	17.6	1,523,281
2026 H1	Native	42.1	50.2	2,024,310
	Stablecoin	41.8	46.0	1,855,019
<i>Panel B. Named currencies in the trading graph</i>				
Currency	Year	Betweenness (rank)	Intermediary-position share [%]	Route-participation share [%]
WETH	2020	0.965 (1)	70.7	77.2
	2024	0.988 (1)	76.2	80.5
	2026 H1	0.925 (1)	42.4	50.7
USDC	2020	0.017 (4)	10.7	11.7
	2024	0.045 (2)	8.1	8.6
	2026 H1	0.107 (3)	21.1	25.1
USDT	2020	0.096 (2)	10.7	11.7
	2024	0.027 (3)	6.8	7.1
	2026 H1	0.113 (2)	16.2	19.3

Note: Panel A uses all coherent complete routes that do not return to their starting asset. Intermediary-position shares use all intermediary positions as the denominator. Route-participation shares use complete routes as the denominator and may sum above 100% across currency families. The 2020 and 2024 rows use full calendar years; 2026 H1 covers January through June. Panel B builds one undirected trading graph for each year from legs observed on the fifteenth day of each month. An edge records at least one leg in either direction during the sampled year. Betweenness is unweighted and approximated from 256 source nodes with a fixed seed. Realised-use shares in Panel B use the same sampled dates and retain all route lengths. Ranks are shown in parentheses.

H Secondary route and liquidity comparisons

This section reports evidence used directly to interpret the main estimates: entry-activity restrictions, differences across venue pricing families, the composition of capital growth before first use, and subsequent comovement between route use and capital. Pool states and first-use dates are equilibrium outcomes; the estimates describe how route use and liquidity line up.

H.1 Persistence among pairs with greater entry activity

Table 23 repeats the post-entry stablecoin-share regressions after requiring at least five or ten routes on the entry day. The estimates remain close to one-for-one at both horizons. The main persistence result is also present among entrants with a more precisely measured initial allocation.

H.2 Venue pricing families

Table 24 divides routes by the pricing family used on every leg. The stablecoin rotation is present among routes priced entirely by the constant-product family, which carries 84.8% of vehicle-currency intermediary positions in 2026.

H.3 Capital composition and later adjustment

Table 25 separates newly active pools from capital growth in pools already supporting the route. Newly active pools hold 7.5% of stablecoin weak-leg capital on the first-use day; 92.5% lies in pools active one week earlier. Table 26 follows the same local bridge across exact 30-, 60-, and 120-day horizons. Under route-activity weights, current depth is positively associated with later route share. Equal pair-date-scope weights yield smaller forward estimates. The time-reversed benchmarks are also positive in some specifications, especially under equal weights. The table therefore records persistent comovement; provider response to route use remains unresolved.

H.4 Relative-price risk and deposited bridge capital

Lower endpoint-vehicle relative-price risk is associated with deeper constant-product bridges at the measurement date and 30 days later. This relation is consistent with divergence risk influencing where liquidity providers place capital. Panel B of Table 27 gives the corresponding vehicle comparison. Stablecoin bridges have higher median risk than WETH bridges and carry the lower relative volatility in fewer than one third of comparable pair-months. Divergence risk can therefore help locate bridge capital without accounting for the aggregate rise of stablecoin intermediation.

Table 23: Post-entry persistence among pairs with greater entry activity

<i>Stablecoin route share among entrants with greater first-day activity</i>				
Minimum first-day routes	Days 1–30		Days 31–120	
	5	10	5	10
Entry stablecoin share [effect pp per 10 pp]	+9.70*** (0.19)	+9.65*** (0.32)	+9.31*** (0.44)	+9.35*** (0.60)
Observations (retrading pairs)	1,629	671	885	452
Eligible entrants	2,192	775	2,192	775
Entry-date clusters	115	93	100	80
Pair controls and equal-pair weights	Yes			

Note: Each column regresses the stablecoin share in the stated later window on the stablecoin share observed on the entry day. The sample requires complete follow-up through day 120 and subsequent trading in the stated window; the entry day is excluded from every later outcome. All columns control for cohort year, endpoint class, log entry routes, direct-route share, and longer-route share and use equal-pair weights. Coefficients and entry-date-clustered CR1 standard errors in parentheses report the outcome change in pp for a 10 pp higher entry share. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 24: Vehicle role specialisation by venue pricing family

Year	All venues		Constant product		Balancer		Curve	
	Count	Value	Count	Value	Count	Value	Count	Value
Panel A: Stablecoins								
2020	1.45	0.42	1.50	0.42	no routes		no intermediation	
2021	1.18	0.89	1.20	0.92	2.09	1.43	no intermediation	
2022	1.46	1.08	1.50	1.12	1.91	1.00	no intermediation	
2023	1.89	0.94	1.95	1.00	1.46	0.37	no intermediation	
2024	1.27	0.84	1.28	0.78	1.01	0.27	no intermediation	
2025	1.23	1.21	1.23	1.20	0.70	0.09	no intermediation	
2026	1.41	1.24	1.42	1.33	1.75	1.68	no intermediation	
Panel B: Native asset								
2020	0.91	1.56	0.90	1.37	no routes		no intermediation	
2021	0.94	1.13	0.94	1.09	0.24	0.66	no intermediation	
2022	0.85	0.96	0.85	0.93	0.35	0.90	no intermediation	
2023	0.87	1.16	0.88	1.08	0.50	0.53	no intermediation	
2024	0.95	1.20	0.95	1.29	0.22	0.20	no intermediation	
2025	0.89	0.67	0.92	0.85	0.19	0.19	no intermediation	
2026	0.77	0.58	0.80	0.62	0.51	1.30	no intermediation	

Note: Role specialisation is an asset type's share of intermediary activity divided by its share of route-endpoint demand, formed within each venue family. A route enters a family only when every leg uses a venue in that family. The constant-product family contains Uniswap v2 through v4 and SushiSwap v2 and v3. Uniswap v1 remains in the full route panel but is reported separately in this venue-family architecture comparison. Balancer is listed separately because its pools use both weighted and stable invariants. Values use all routes with positive dollar support. The ratio measures vehicle use relative to endpoint use; it is distinct from aggregate vehicle share.

Table 25: Pool activation and capital growth before stablecoin route use

Outcome	Stablecoin	WETH	Difference
Weak-leg deposited capital [log change]	+0.76*** (0.16)	+0.06 (0.07)	+0.70*** (0.17)
Active pool count [log change]	+0.04*** (0.01)	-0.01* (0.01)	+0.05*** (0.01)
Any newly active pool [pp]	+7.32*** (1.66)	+0.41 (0.41)	+6.91*** (1.72)
New-pool share of adoption capital [pp]	+7.47*** (1.70)	+0.41 (0.41)	+7.14*** (1.77)
Continuing-pool capital [log change]	+0.23*** (0.07)	+0.07* (0.04)	+0.16** (0.08)

Note: The sample contains the bridge events in Figure 2. Stablecoin and WETH outcomes use each vehicle's weaker leg. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; a continuing pool has positive capital on both dates. Each cell is an equal-event mean. Parentheses contain standard errors two-way clustered by pair and adoption date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 26: Route use and local bridge depth at later horizons

Conditional level equation	30 days	60 days	120 days
<i>Panel A: Forward levels, route-activity weights</i>			
$R_{b,t} \rightarrow B_{b,t+h}$, pooled [log points]	-0.007* (0.004)	-0.007 (0.006)	+0.010 (0.014)
$R_{b,t} \rightarrow B_{b,t+h}$, stablecoin [log points]	-0.002 (0.007)	-0.002 (0.005)	+0.017 (0.012)
$B_{b,t} \rightarrow R_{b,t+h}$, pooled [pp]	+0.728*** (0.239)	+1.665*** (0.463)	+3.281** (1.315)
$B_{b,t} \rightarrow R_{b,t+h}$, stablecoin [pp]	+1.823* (0.938)	+4.837*** (1.718)	+4.916** (2.099)
<i>Panel B: Time-reversed benchmark, route-activity weights</i>			
$R_{b,t+h} \rightarrow B_{b,t}$, pooled [log points]	+0.012*** (0.004)	+0.029*** (0.009)	+0.026** (0.011)
$R_{b,t+h} \rightarrow B_{b,t}$, stablecoin [log points]	+0.015** (0.006)	+0.032*** (0.008)	+0.036*** (0.010)
$B_{b,t+h} \rightarrow R_{b,t}$, pooled [pp]	-0.483 (0.323)	-0.339 (0.343)	+1.114 (1.600)
$B_{b,t+h} \rightarrow R_{b,t}$, stablecoin [pp]	-0.599 (1.073)	+0.065 (0.806)	+2.930 (1.990)
<i>Panel C: Forward levels, equal pair-date-scope weights</i>			
$R_{b,t} \rightarrow B_{b,t+h}$, pooled [log points]	+0.000 (0.001)	-0.000 (0.002)	+0.006 (0.005)
$R_{b,t} \rightarrow B_{b,t+h}$, stablecoin [log points]	+0.002 (0.002)	+0.004 (0.003)	+0.015*** (0.006)
$B_{b,t} \rightarrow R_{b,t+h}$, pooled [pp]	+0.841 (0.565)	+1.274 (0.818)	+0.248 (1.086)
$B_{b,t} \rightarrow R_{b,t+h}$, stablecoin [pp]	+1.117 (0.712)	+2.107** (1.068)	+1.577 (1.165)
<i>Panel D: Time-reversed benchmark, equal pair-date-scope weights</i>			
$R_{b,t+h} \rightarrow B_{b,t}$, pooled [log points]	+0.003* (0.002)	+0.006* (0.003)	-0.001 (0.004)
$R_{b,t+h} \rightarrow B_{b,t}$, stablecoin [log points]	+0.003 (0.002)	+0.008** (0.004)	+0.004 (0.005)
$B_{b,t+h} \rightarrow R_{b,t}$, pooled [pp]	+0.646 (0.604)	+0.320 (0.865)	+1.015 (1.185)
$B_{b,t+h} \rightarrow R_{b,t}$, stablecoin [pp]	+1.431** (0.629)	+1.715* (1.020)	+3.592*** (0.990)
Pooled rows / pair clusters / dates	51,204 / 502 / 303	39,964 / 435 / 243	18,303 / 339 / 123
Stablecoin rows / pair clusters / dates	30,181 / 483 / 303	23,389 / 421 / 243	10,608 / 326 / 123
Bridge and date effects			Yes
Initial outcome level			Cubic
Two-way clustered covariance			Pair and date

Note: For source–vehicle–destination bridge b , $R_{b,t}$ is one vehicle’s share of route activity among WETH, WBTC, DAI, USDC, and USDT, and $B_{b,t} = \log(1 + \min\{K_{iv,t}, K_{vj,t}\})$ is weak-leg deposited capital. Stablecoin rows retain DAI, USDC, and USDT separately. The sample draws on January–June 2024 and January–June 2026 and retains continuing bridges with positive deposited capital and positive five-vehicle route activity at both exact dates. Forward panels regress the exact future level on the dated predictor, a cubic in the current outcome level, bridge effects, and date effects. Time-reversed panels use the end-of-window predictor for the earlier outcome and a cubic in the end-of-window outcome level. Route-activity weights are measured on the predictor date; equal weights make each pair-date-scope sum to one. Route-use rows report the coefficient for 10 pp more route share in natural-log points; bridge-depth rows report the coefficient for one natural-log point more depth in pp. Standard errors are two-way clustered by pair and date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The p-values are unadjusted for the displayed family of horizons, samples, timings, and weights.

Table 27: Relative-price risk and deposited bridge capital

Log deposited bridge capital	(1) Depth at t	(2) Depth at t	(3) Depth at $t + 30$	(4) Depth at $t + 30$
<i>Panel A: Relative-price risk and bridge depth</i>				
Relative volatility [10 pp]	-0.117** (0.049)		-0.093*** (0.029)	
Daily divergence loss [1 bp]		-0.053** (0.021)		-0.044** (0.020)
Log prior vehicle-route count	+2.218*** (0.097)	+2.235*** (0.095)	+0.613*** (0.066)	+0.625*** (0.067)
Log initial bridge depth			+0.733*** (0.022)	+0.734*** (0.022)
Within R^2	0.322	0.321	0.702	0.702
Observations	58,447	58,447	58,447	58,447
Pair clusters	4,254	4,254	4,254	4,254
Date clusters	71	71	71	71
Pair $\times$ date fixed effects	Yes	Yes	Yes	Yes
Vehicle fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date
<i>Panel B: Native and stablecoin bridge risk</i>				
	Relative volatility		Daily divergence loss	
Native median	126.9%		5.60 bp	
Stablecoin median	148.4%		7.68 bp	
Stablecoin has lower risk [pair-months]	29.4%		28.6%	

Note: The unit is a pair, monthly date, and vehicle among WETH, DAI, USDC, and USDT. Each row requires at least ten routes for the pair during the preceding calendar month, at least 20 daily return observations during days -30 through -1, and positive deposited capital on both route legs for at least one native and one stablecoin bridge in the pair-month. Bridge depth is the smaller deposited-capital stock across the two legs in full-range Uniswap v2 and SushiSwap v2 pools, measured at the prior-calendar state. Relative volatility is the larger annualised endpoint-vehicle volatility across the two legs. Daily divergence loss applies the exact 50-50 constant-product expression to the same relative returns and takes the larger leg value. Columns 3 and 4 control for initial bridge depth; every column controls for prior vehicle-route activity and absorbs pair-by-date and vehicle effects. Standard errors are two-way clustered by pair and date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates describe equilibrium liquidity allocation; fee income, provider hedges, and provider-level returns are unavailable.

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